

Quantifying Fit of Wearable Systems With a Multi-Band, Remote Measurement System

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Abstract

Garment fit is critical in wearable systems and functional clothing, including masks and other personal protective equipment (PPE), and yet no methods exist that can accurately and in real-time quantify their fit characteristics of non-translucent or non-form fitting garments. We propose a technique for remotely measuring the gap between an external wearable device or clothing surface and the underlying, visually-obstructed body surface – which we refer to as the garment-body “air gap” – that commonly occurs in positive ease garments (i.e., garments that are larger in dimension than the underlying body dimension). To do this, we developed a triple frequency band remote measurement system that is based on a millimeter wave radar, an ultrasound sensor, and an infrared distance sensor, which when used synergistically allows for remote measurement of the uniaxial distances to multiple layered surfaces simultaneously. Here this novel, first-generation system is tested and evaluated for the purpose of measuring garment fit.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing design and evaluation methods**; *Ubiquitous and mobile computing systems and tools*.

Keywords

ubiquitous computing, garment fit, wearable technology, mmWave

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1 Introduction

Wearable systems and functional clothing, including masks and other personal protective equipment, PPE, stand to benefit greatly from better methods of garment fit. For most garments, namely, those that are non-translucent and non-form fitting, no methods exist that can accurately and in real-time quantify their fit characteristics. [1, 3–5]

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Wearable systems are often composed of sensors and actuators that depend on garment fit to be effective. Recent advances in measuring garment fit are susceptible to weaknesses that our system attempts to overcome by 1) being a remote, non-garment integrated measurement system and 2) by taking advantage of the fact that different measurement technologies at different wavelengths (i.e. infrared vs. ultrasonic vs. millimeter wave) reflect off objects in different and distinguishable ways.

Among remote measurement systems used for garment fit, one limitation our multi-band system overcomes handily: the inability to measure *through* clothing.

Millimeter wave (mmWave) frequency modulated, continuous wave (FMCW) radar has the ability to measure distances between surfaces, even surfaces that are visually occluded, in our case, the body surface while clothed. The measured distance between the garment and the body surface is what we refer to as an “air gap”. Any substantive air gap that is found through a scan is a measure of the garment’s lack of fit.

2 System Design

Scanning technologies common in surface measurements, infrared (IR) and ultrasound (US), are employed coupled with a mmWave radar for purpose of measuring distance to and *through* garments and wearables.

The triple frequency band remote measurement system is based on a mmWave 77-81 GHz FMCW radar, 40 kHz ultrasound sensor, and an infrared 344 GHz distance sensor, shown in Figure 3. These three sensors working at distinct wavelengths, when used synergistically, allows for remote measurement of the uniaxial distances to multiple layered surfaces simultaneously.

2.1 Enhancing Accuracy with Signal Processing

The mmWave FMCW radar operating at frequencies 77-81 GHz by default measures with only a resolution on the order of centimeters. The application of measuring garment fit however requires millimeters or smaller. To overcome this limitation, our system employs further signal processing on the mmWave radar measurements in the form of the ZoomFFT (zFFT) signal processing algorithm.

The ZoomFFT algorithm requires a window of distances on which to “zoom” into and works essentially like a band-pass filter, allowing the measurement system to focus on a particular depth beyond the garment, precisely to the level of interest, to that of the garment-body “air gap”.

Due to the multi-band nature of the system, we can set the desired depth based it on the distance measurements of the first surface from the other sensor technologies, infrared and ultrasonic.

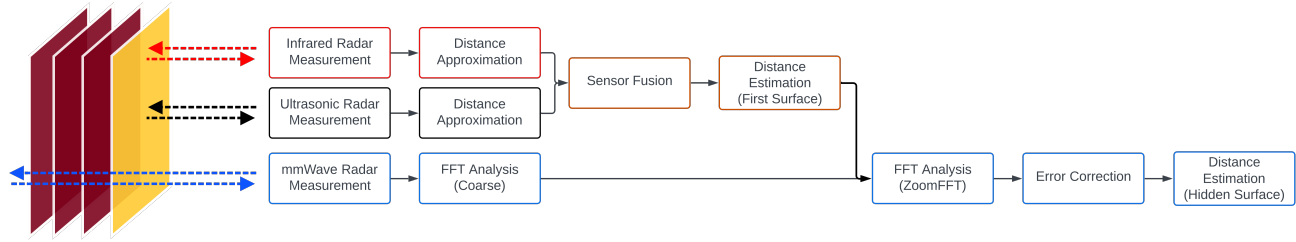


Figure 1: The infrared and ultrasonic sensors measure distances of the “first surface”, typically clothing, while the mmWave radar measures distances of the visually occluded surfaces, the body and also other intervening surfaces.

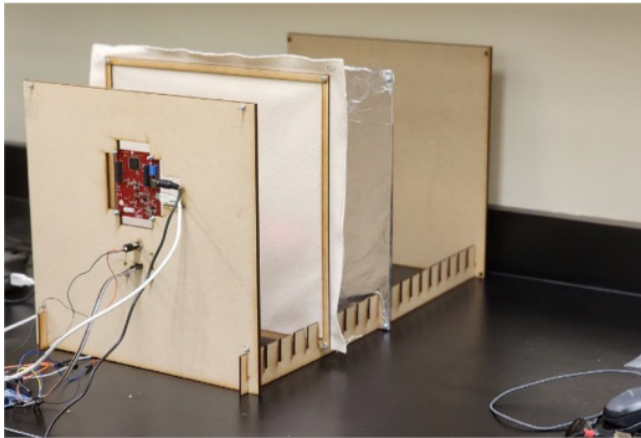


Figure 2: The sensor suite simultaneously measuring distances between two surfaces, the first imitating clothing and the second imitating skin.

2.2 Enhancing Accuracy with Machine Learning

The infrared and ultrasonic sensors’ measurement error can be greatly reduced by way of machine learning, namely, by training a model (supervised learning) using the sensors’ measurements and the actual known distances of the object being scanned. In this way, the IR and US sensors become “fused” to create a much more accurate IR/US measurement of the garment surface.

Infrared and ultrasonic do not penetrate to the visually occluded body surface, and so, our enhancement of the IR/US signal is an enhancement on the accuracy of the whereabouts of the clothing surface only whereas our enhancements with the application of ZoomFFT enhance only the measurements of the body surface. Together, we aim to enhance the accuracy and capabilities of the system for measuring the garment-body “air gap”.

Our optimizations of the system for the purpose of synergistic measuring of garment fit are presented visually in Figure 1.

3 Methodology & Evaluation

The capability of the system to measure garment-body “air gaps” depends on the systems ability to create accurate results across various distances and through a variety of clothing materials. We explore these two possible confounds to accuracy in the following sections.

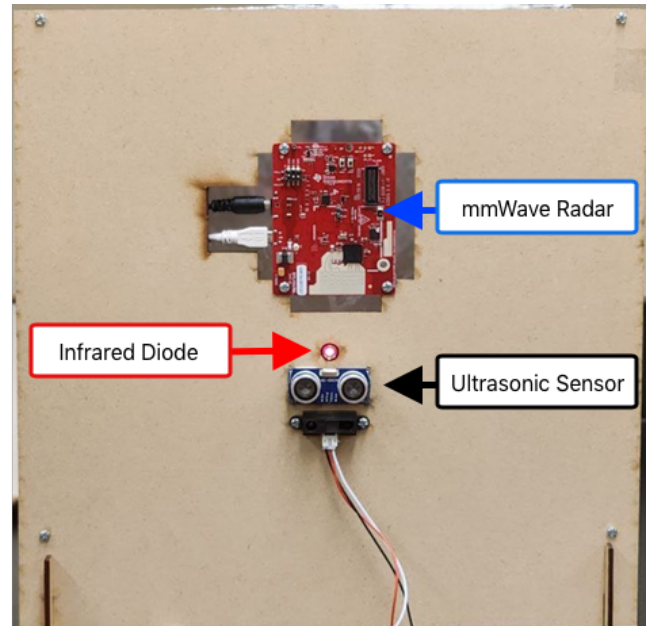


Figure 3: The sensor suite containing, from top to bottom, the mmWave radar, the infrared diode, the ultrasonic distance sensor, and the infrared distance sensor.

3.1 Accuracy and Distance

Our measurement system was tested for accuracy with a single surface at various distances from the sensor suite, from 6 to 60 cm at 3 cm intervals. The sensor suite is pictured in Figure 2. Three measurements were taken at each distance point and averaged together. The measured surface was aluminum foil since it has been shown to have a similar reflectance to human skin.[2]. The errors by sensor technology at various distances are shown in Figure 4.

As shown, the inaccuracies for the scanning technologies are on the order of millimeters, suitable for our application of measuring garment fit.

3.2 Enhancements with Machine Learning and ZoomFFT

Our machine learning strategy increased the accuracy of the system from 0.1610 cm and 0.4728 cm in measurement error for US and

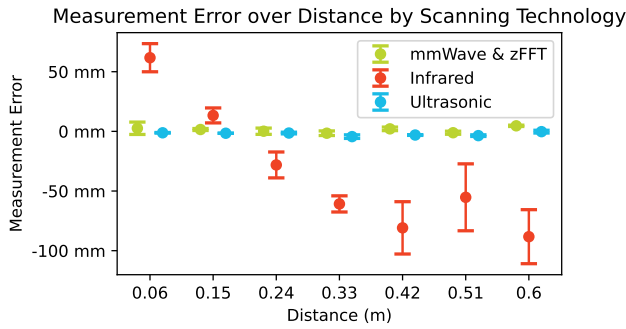


Figure 4: The accuracy of each sensor technology on a single surface over various distances.

Table 1: Various materials are scanned and compared to the reflectivity of the body surface.

Material	Reflectivity Index
Aluminum Foil	1
Denim	0.6667
Cotton Duck Canvas	0.381
Polypropylene	0.0024

IR sensors, respectively, to 0.41 mm of error for the IR/US sensor fusion measurements.

Similarly, though the application of the ZoomFFT algorithm for further signal processing on measurements of the mock-body surface, errors were reduced from 1.1 cm to 4.4 mm RMSE.

3.3 Accuracy and Materials

Across the electromagnetic spectrum, accuracy of our distance measurements depended greatly on the response of the materials involved in the scan. Materials serving as a mock-clothing vary in their optical and sonic response. Further, we examined various textiles and their effect on the system’s ability to measure the distance to the first surface, and the first surface’s potentially confounding effect on accuracy of second surface measurements.

We found that our ability to discern “air gaps” relies on surfaces to have sufficiently different responses under the mmWave radar. If their responses are too similar when they are close together, our system is largely unable to discern which is the second surface and which is the first.

A “reflectivity index” is useful then when characterizing various materials placed close together and predicting the measurement accuracy we can expect from the mmWave sensor response.

Here the aluminum foil / skin-like surface serves as our baseline with a reflectivity index of 1. Our reflectivity index is based each surface’s mmWave range profile, a measurement of how each material responds under the mmWave radar. See Table 1 for our calculated reflectivity indices for the various garment materials we tested.

Given sufficient distance between the materials, two materials with a similar reflectance index can be discerned using the

Table 2: Strong reflective differences between materials and skin lead to lower errors in measurement of fit via the FMCW radar.

Material	Reflectivity Diff. with Skin	Error (m)
Denim	0.3333	0.0073
Cotton Duck Canvas	0.619	0.013
Polypropylene	0.9976	0.0019

zoomFFT’s “zoom window” to further discriminate between the two materials.

Each material was scanned with the sensor suite and errors were recorded along side the reflectivity index difference between the mock clothing material and the mock skin (aluminum foil) surface behind it.

Table 2 captures the results which show that measurement accuracy is not dependent on only properties of the clothing surface alone, but also on discernible differences in electromagnetic response between the surfaces.

4 Conclusion

We believe our first-generation measurement system described here represents a promising and novel technique for measuring garment fit to ensure the effectiveness of functional clothing, PPE, and a variety of wearable systems.

5 Future work

Our system is, so far, limited to doing uniaxial measurements, like single pixels of a garment and the body beneath it. The enterprise of measuring fit for a garment requires measurements from all angles. Our future work includes extending the system to bi-axial measurements to create a visualization and analysis similar to that of a 3D body scan, with the key advantage over a traditional 3D body scan being the ability to scan the clothing as well as the body surface beneath it in a single scan.

Future work also includes further accuracy enhancements via machine learning, specifically using additional information derived from the textile surfaces themselves, with reflectivity index described above being one such piece of additional information that can enhance our system’s accuracy.

A remote, real-time garment fit measurement system such as ours also affords measuring garment fit for functional clothing and PPE while the body is in motion.

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References

- [1] Simeon Gill. 2011. Improving garment fit and function through ease quantification. *Journal of Fashion Marketing and Management: An International Journal* 15, 2 (2011), 228–241. Publisher: Emerald Group Publishing Limited.
- [2] Jaime Laviada, Ana Arboleya-Arboleya, Yuri Álvarez, Borja González-Valdés, and Fernando Las-Heras. 2017. Multiview three-dimensional reconstruction

- by millimetre-wave portable camera. *Scientific Reports* 7, 1 (July 2017), 6479. <https://doi.org/10.1038/s41598-017-06475-7>
- [3] Simon Ozbek, Md Tahmidul Islam Molla, Crystal Compton, and Brad Holschuh. 2018. Novel manufacturing of advanced smart garments: knitting with spatially-varying, multi-material monofilament. In *Proceedings of the 2018 ACM International Symposium on Wearable Computers*. 120–127.
- [4] Hyewon Seo, See-Jo Kim, Frederic Cordier, and Kyunghi Hong. 2007. Validating a cloth simulator for measuring tight-fit clothing pressure. In *Proceedings of the 2007 ACM symposium on Solid and physical modeling*. 431–437.
- [5] Olga Troynikov and Elnaz Ashayeri. 2011. 3D body scanning method for close-fitting garments in sport and medical applications. In *HFESA 47th annual conference*. Crows Crows Nest, 11–16.